

# Check fitted flocker model

Specify model to check:

```
spec <- params$spec #final_species_eco_sorted[6] # 1
spec
```

```
[1] "Bell's Vireo"
```

```
buffer_km <- params$buffer_km
buffer_km
```

```
[1] 750
```

```
fold <- params$fold
# fold

results_dir <- params$results_dir
results_dir
```

```
[1] "M:/Documents/DEBTs/analysis/Schifferle_BBS_occupancy_models_2023/results/full_model"
```

Load fitted model:

```
load(file = file.path(results_dir,
                       paste0("out_", spec, "_fm_buffer", buffer_km, ".RData"))) # xx adjust
#print(out)

posterior <- brms::as_draws(out)
```

**Check whether MCMC worked:**

**Do we have divergent transitions?**

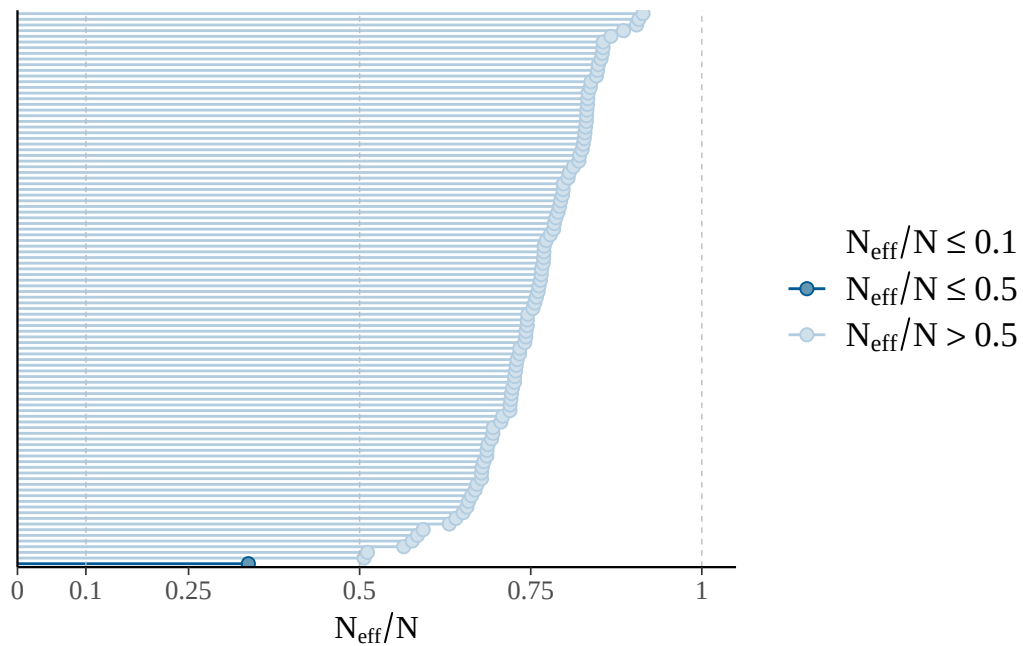
```
[1] "divergent transitions: 0"
```

**Explorations regarding divergent transitions:**

...

**Is effective number of MCMC samples large enough:**

The larger the ratio of  $n_{\text{eff}}$  to  $N$  (number of samples), the better.



```
[1] "effective sample size fine"
```

```
[1] "n_eff ratio bulk ESS:"
```

| Min.   | 1st Qu. | Median | Mean   | 3rd Qu. | Max.   |
|--------|---------|--------|--------|---------|--------|
| 0.6715 | 0.9349  | 1.0924 | 1.1563 | 1.3088  | 2.3206 |

```
[1] "n_eff ratio tail ESS:"
```

| Min.   | 1st Qu. | Median | Mean   | 3rd Qu. | Max.   |
|--------|---------|--------|--------|---------|--------|
| 0.5064 | 0.7218  | 0.7656 | 0.7604 | 0.8270  | 0.9140 |

```
[1] "effective sample size in bulk and tail fine"
```

**Do chains mix well:**

No divergences to plot.

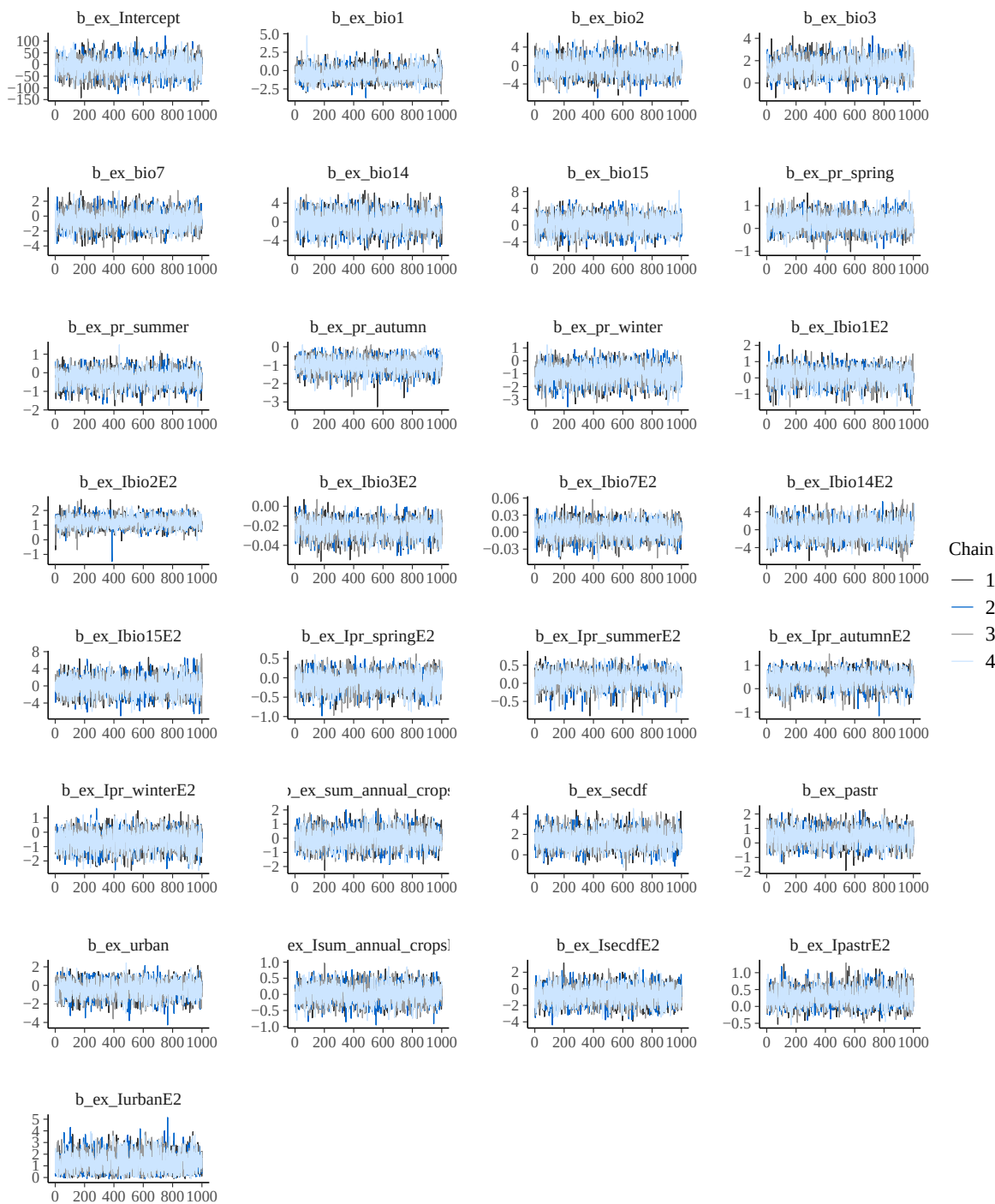
No divergences to plot.



No divergences to plot.



Post-warmup iteration



Post-warmup iteration

### Are R-hat values fine:

| Min.   | 1st Qu. | Median | Mean   | 3rd Qu. | Max.   |
|--------|---------|--------|--------|---------|--------|
| 0.9991 | 0.9996  | 0.9998 | 1.0000 | 1.0002  | 1.0017 |

```
[1] "All R-hat values < 1.01"
```

### Print model posteriors

```
[1] "parameters for which uncertainty interval doesn't overlap zero:"
```

```
[1] "b_occ_bio1_3yrs"          "b_occ_Ibio7_3yrsE2"
```

```
[3] "b_occ_Ipr_spring_3yrsE2" "b_occ_Ipastr_3yrsE2"
```

```
[1] "b_colo_bio1"              "b_colo_pr_winter"
```

```
[3] "b_colo_Ipr_winterE2"      "b_colo_sum_annual_crops"
```

```
[5] "b_colo_secdf"             "b_colo_Isum_annual_cropsE2"
```

```
[1] "b_ex_pr_autumn" "b_ex_Ibio2E2"  "b_ex_Ibio3E2"  "b_ex_IurbanE2"
```

